CURE

Dynamic Memory Curation via Retroactive Eviction for Test-Time Out-of-Distribution Detection

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Problem, Related Work, Pipeline and Data

Outline

1. Introduction and Problem Statement
2. Related Work
3. Gaps and Limitations
4. Proposed Method: CURE
5. Pipeline
6. Data

1.1 Background

- Machine learning models are deployed in image classification, network monitoring, environmental sensing and diagnosis support
- In deployment they meet inputs never seen in training: **out-of-distribution (OOD)** data
- The risk: on unfamiliar inputs, a model still answers, often with high confidence

1.2 Evidence of Need

- An MNIST classifier gives pure random noise 91% mean confidence - Hendrycks & Gimpel, ICLR 2017
- On ImageNet-O (2,000 natural images of unseen classes), ResNet-50 is near chance at telling OOD from ID - Hendrycks et al., CVPR 2021
- The first toxic Dinophysis bloom in the U.S. (2008) was caught early only by a continuously running imaging system; shellfish harvesting was halted - Campbell et al., J. Phycol. 2010
- When real events are rare, even a small false-alarm rate means most alarms are false, and operators stop trusting the system - Axelsson, ACM TISSEC 2000
- Self-updating systems can be pulled toward the data they accept: adaptive biometric templates, online centroid detectors - Biggio et al., S+SSPR 2012 and Kloft & Laskov, JMLR 2012

> **Two needs at once:** detect novelty, and keep false alarms low

1.3 Problem Statement

- Test data arrive as a continuous, unlabeled stream
- Remembering past OOD samples could help catch similar ones later
- But an ID sample stored as OOD makes similar ID look OOD, and they get stored too: **self-poisoning**
- Strict admission keeps the memory clean, but nearly empty

> **Question:** can memory admit freely, then evict its mistakes using later data, without labels?

1.4 Objectives and Scope

- Survey OOD detection methods, focusing on test-time memory
- Identify the shared limitations of these methods
- Propose **CURE**: label-free memory curation where every entry expires unless later data confirm it
- Scope: no test labels, no retraining, data processed in stream order

2.1 Related Work: Overview

52 papers in 7 groups

OOD detection and test-time methods

1. Static OOD detectors (10)
2. VLM and cache (4)
3. Test-time adaptation (6)
4. Dynamic-memory OOD (7)

Ideas and tools CURE builds on

5. Retention and buffer cleaning (11)
6. Self-poisoning and defenses (6)
7. Statistical tests (8)

2.1 Group 1: Static OOD Detectors

Method	Idea	Limitation
MSP, ODIN, Energy, GEN	Classifier confidence as the score	Still confident on OOD
Mahalanobis, kNN, ViM	Feature distance to ID data	Depends on feature quality
ReAct, ASH, SEM	Prune abnormal activations	Tuning varies by backbone

Shared gap: Each sample is scored alone; weak on near-OOD

2.1 Group 2: VLM and Cache

Method	Idea	Limitation
MCM, NegLabel	CLIP image–label similarity, plus negative labels	Depends on CLIP and prompt design
TDA, DMN	Cache of confident test samples	Entries replaced at admission, never reviewed

Shared gap: Built for classification, not novelty detection

2.1 Group 3: Test-Time Adaptation

Method	Idea	Limitation
TENT, EATA, SAR	Update normalization layers by minimizing entropy on test data	The self-training loop can collapse
CoTTA, RoTTA, ATTA	Mean teacher, class- and age-balanced memory	Memory contamination by OOD not studied

Shared gap: assumes the stream holds only known classes

2.1 Group 4: Dynamic-Memory OOD (1/2)

Method	Idea	Limitation
OODD	Fixed ID dictionary plus an OOD queue	A large queue accumulates ID samples
DCAC	Class-wise cache, first in first out	The gate leaks ID, never measured
DynProto	OOD prototypes per class, evicted by age	Worse than static when OOD is rare
TTL	Learns OOD text prompts at test time	Can get worse on near-OOD

Common pattern: store suspected OOD, then score later samples with it

2.1 Group 4: Dynamic-Memory OOD (2/2)

Method	Idea	Limitation
AdaODD	Admits confident ID and OOD, never evicts	Threshold-sensitive, no way to correct errors
WARDEN	Certified e-BH gate at admission	Large TPR loss when OOD is rare
RTL, RTL++	Per-batch linear refit of OOD scores	Needs enough OOD in every batch

Shared gap: no method re-checks an admitted entry using later data

2.1 Group 5: Retention and Buffer Cleaning

Method	Idea	Limitation
DenStream, CluStream, MINAS	Keep only clusters reinforced by new data	Hand-set thresholds, no statistical test
COMPOSE	Labels follow slowly drifting clusters	Needs initial labels, slow drift only
SPR, Co-teaching, SRR, LatentOE	Clean stored samples before reuse	Needs labels or training
LOF, hubness	Density relative to neighbors	Static; hubs bias neighbor counts

Takeaway: review after storage exists, but not for unlabeled OOD

2.1 Group 6: Self-Poisoning and Defenses

Method	Idea	Limitation
Kloft and Laskov, Biggio et al.	An attacker drags a self-updating detector	Deliberate attacks, not natural drift
Certified kNN, ATOM, PoisonedRAG	Robustness of reference stores to poisoning	Training or retrieval setting, not OOD memory

Takeaway: a self-updated reference memory is an attack surface

2.1 Group 7: Statistical Tests Used in CURE

Method	Idea	Limitation
Conformal p-values	Rank a score against clean calibration data	Needs exchangeable calibration data
e-BH, LORD, e-LOND	Control the false discovery rate (FDR)	e-values are more conservative than BH
Cauchy combination, testing randomness	Merge dependent p-values	Accurate only in the tail

Takeaway: The tools behind CURE's keep-or-evict decision

2.2 Current State of the Field

- **Static detectors** are mature but plateauing: no method wins on every benchmark, and near-OOD stays hard - Zhang et al., DMLR 2024
- **Dynamic memory** scores high, but no method re-checks a stored entry using later data
- **The safe route** (WARDEN) blocks admission with a statistical test, and the memory starves when OOD is rare
- **Review after storage** exists in stream clustering and buffer cleaning, but not for unlabeled OOD memory

3.1 Research Gaps

1. Entries kept by their admission score pile up extreme and stale samples
2. Strict admission starves the memory
3. No quantitative law linking drift, memory length and error growth
4. No principled rule for when to stop trusting the memory
5. No valid statistical test for reviewing stored entries
6. No evaluation on time-ordered streams; contamination inside memory is never measured

3.2 Shared Limitations of SOTA

Approach	Examples	Outcome
Open admission	OODD, DCAC, DynProto	Self-poisoning
Strict gating	WARDEN	Memory starves
Admit, then review	None yet	Gap for CURE

Also open: Under which data conditions does memory help, and when does it hurt?

4.1 Idea (1/2)

- Label-free, no retraining; works on features of a frozen backbone
- Every test sample first waits in a quarantine queue
- After d later samples, each sample is reviewed using the samples that came after it
- A denser OOD cluster around it than the ID tail can produce → certify into the curated dynamic memory (CDM); otherwise → evict
- Only samples in the CDM are used for scoring; while the CDM is empty, CURE returns the static score

4.1 Idea (2/2): Design Principles

Open admission, forget by default	Every suspect is reviewed; none is trusted without evidence (gaps 1, 2)
Review with error control	Beta-binomial p-value against the clean ID tail, then e-BH to limit false certifications (gap 5)
Calibrated output with fallback	Conformal memory score, Cauchy fusion with the static score; static score when memory is empty (gap 4)

4.2 Algorithm (1/2)

1. Static score: s_0 = cosine to the K-th nearest ID neighbor (K chosen by APS on validation data), converted to a conformal p-value

2. Screening: samples with $s_0 \leq \tau$ count as evidence; τ = ρ-quantile of s_0 on the calibration set C ($\rho = 20\%$)

3. Quarantine: every sample waits; not used for scoring

4. Review: after d = 256 later samples, count the screened ones within the sample's radius → beta-binomial p-value

4.2 Algorithm (2/2)

5. Certify: p-value → e-value, e-BH (q = 0.2) over samples expiring in the batch; pass → CDM, fail → evict

6. CDM: certified samples only; FIFO, M = 512; every sample in the CDM is used for scoring

7. Memory channel: similarity to the k_m-th nearest sample in the CDM → conformal p-value

8. Final score: Cauchy combination of both p-values; empty CDM → static score

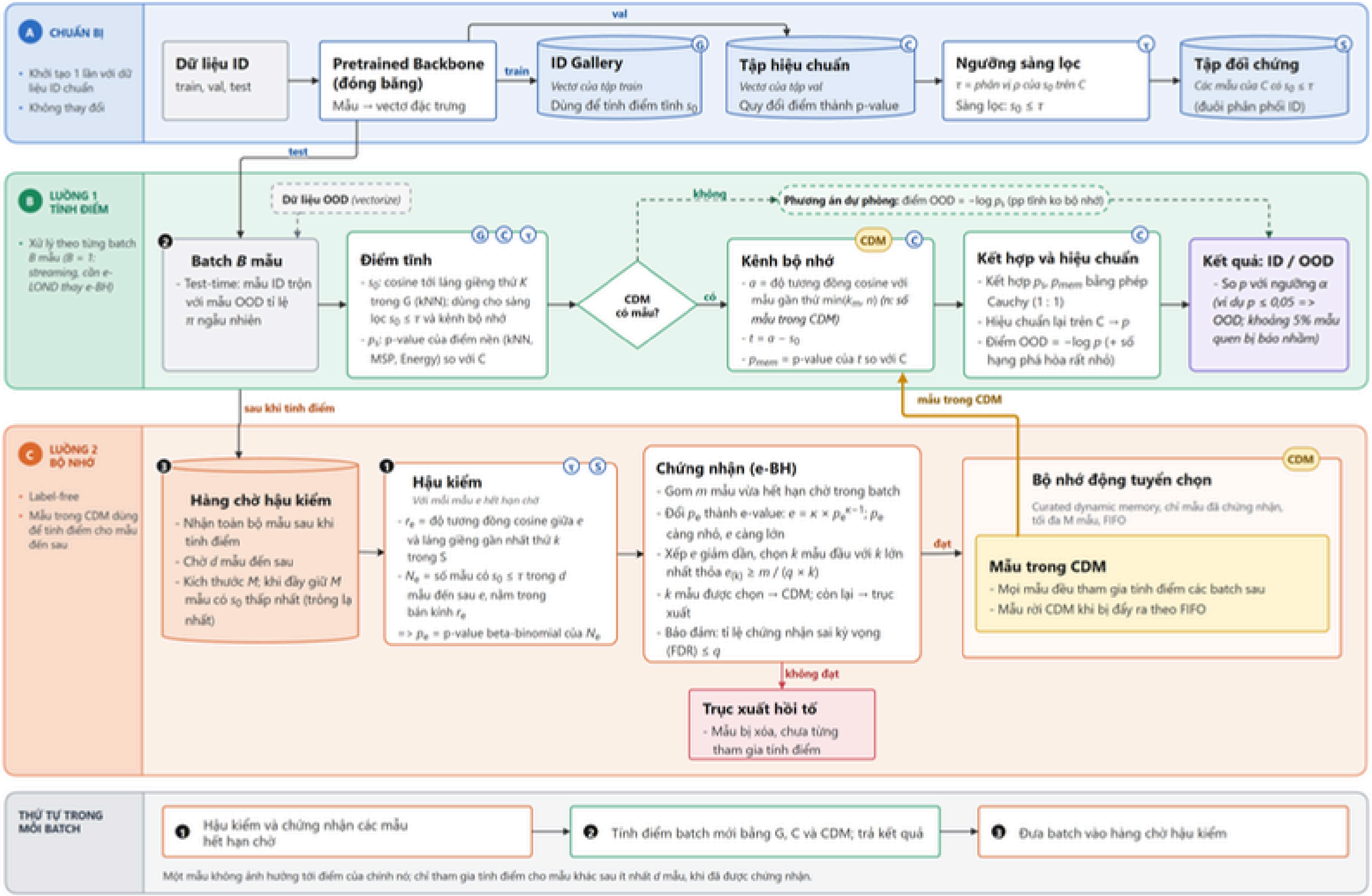
> Causal at batch level (128 samples); stream labels used only for evaluation

5. Pipeline

(A) Offline, ID data only.

(B) Scoring: static score plus the memory channel when the CDM has samples, otherwise the static score.

(C) Memory: every sample waits in the quarantine queue, then e-BH certifies it into the CDM or evicts it.



6.1 Dataset Selection (1/2)

- Data: OpenOOD v1.5, official checkpoints and image lists; 3 ID sets with their official near/far-OOD sets
- Near-OOD: new classes, same image style as ID (overlapping classes removed by OpenOOD); hard
- Far-OOD: different classes and style (digits, house numbers, textures, scenes); easier

ID dataset	Near-OOD	Far-OOD
CIFAR-10	CIFAR-100, TinyImageNet	MNIST, SVHN, Textures, Places365
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ImageNet-200	SSB-hard, NINCO	Textures

6.1 Dataset Selection (2/2)

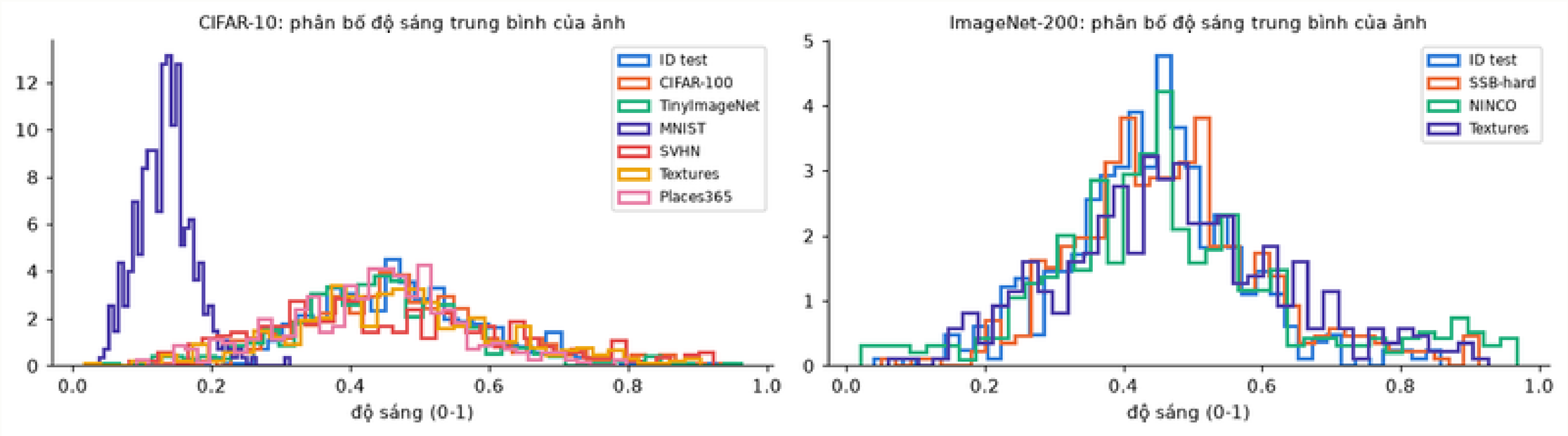
- **CIFAR-10/100:** 50,000 train (gallery), 1,000 val (calibration), 9,000 test
- **ImageNet-200:** ID test split by class; half A → 3,520 gallery + 880 calibration, half B → 4,600 stream
- **Backbones:** ResNet18 on all; SupCon and DINOv2 on CIFAR-10
- OpenImage-O and iNaturalist (ImageNet-200 far-OOD): obtained later, not yet in experiments, so ImageNet-200 far-OOD is Textures only
- id_val both selects K and calibrates, so exchangeability is only approximate; CIFAR-100 has only 10 calibration images per class

6.2 Stream Construction

- One stream = the whole ID test set + n_OOD samples drawn at random from one OOD set, with $n_OOD = n_ID \times \pi / (1 - \pi)$; if the OOD set is too small, ID is reduced instead
- CIFAR-10/100 (9,000 ID test): $\pi = 5\% \rightarrow$ 474 OOD (9,474 samples); 20% → 2,250 (11,250); 50% → 9,000 (18,000)
- ImageNet-200: 4,600 ID test samples (half B) + OOD at the same ratios
- Order shuffled at random, 3 seeds (0, 1, 2): 27 (backbone, OOD set) pairs × 3 π = 81 groups, 243 streams; seeds 3-5 kept for confirmation
- Gallery G and calibration set C never enter the stream; labels are used only for evaluation
- Processed in batches of B = 128 samples

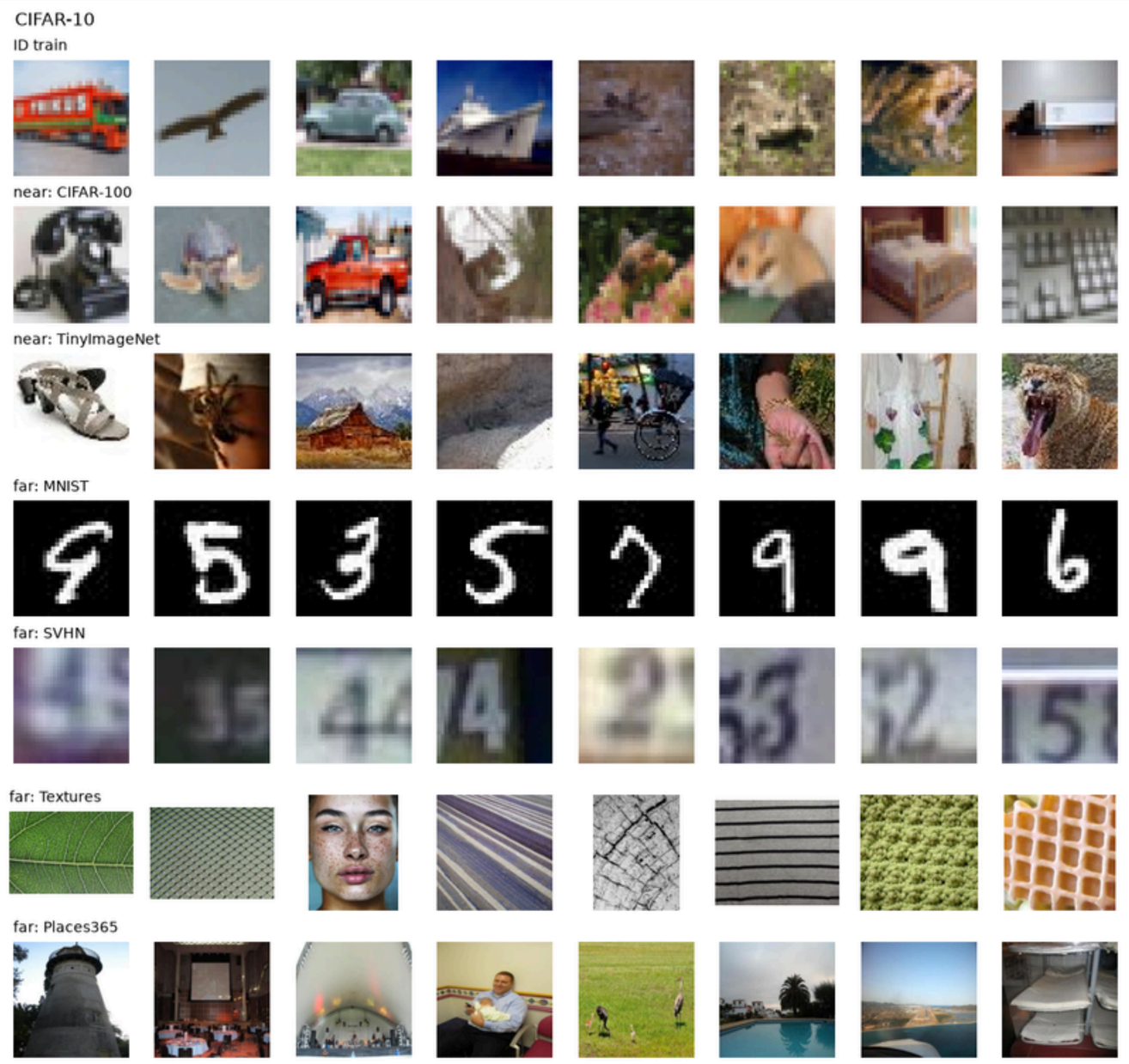
6.3 EDA: Raw Images

- Class balance: all three ID sets are perfectly balanced (e.g. 5,000 train images per class in CIFAR-10)
- Duplicates: no exact duplicates between train and test, or between ID and near-OOD (MD5 on 2,000 images per set)
- Sizes: differ by set (MNIST 28×28 grayscale, TinyImageNet 64×64, Places365 256×256); all resized before the network, but resize and format traces (CIFAR is PNG, OOD sets are JPEG) may make far-OOD easier
- Pixel statistics: near-OOD ≈ ID (mean brightness 0.475 vs 0.471); MNIST is dark and colorless



6.3 EDA: Sample Images (CIFAR-10)

Near-OOD (CIFAR-100, TinyImageNet) looks like ID: natural objects at a similar resolution, only the classes differ. Far-OOD changes both class and style: digits, house numbers, textures, scenes.



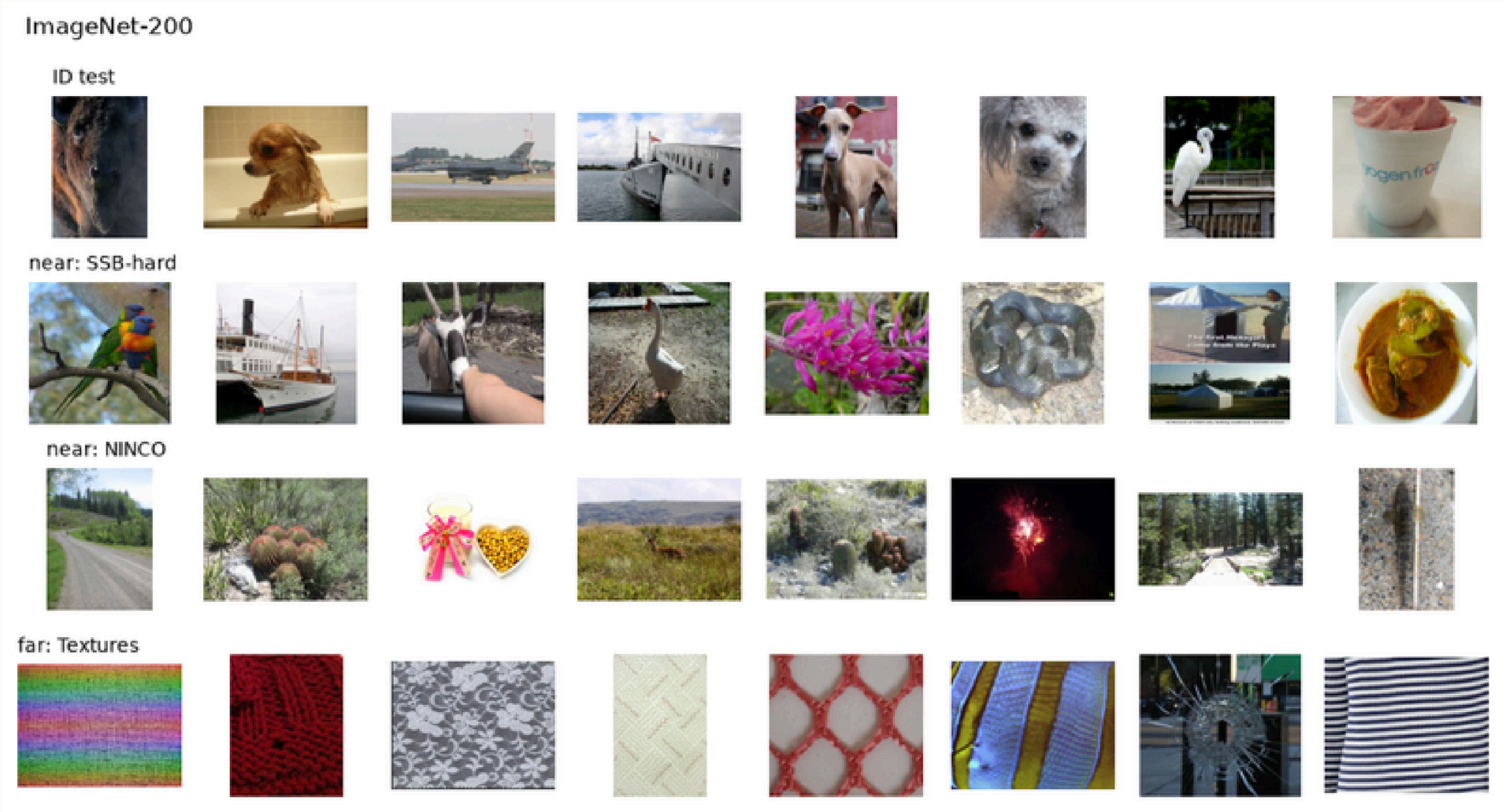
6.3 EDA: Sample Images (CIFAR-100)

Same OOD sets, with CIFAR-10 now near-OOD. With 100 classes and only 10 calibration images per class, CIFAR-100 is the hardest CIFAR benchmark for kNN (AUROC 77-84).



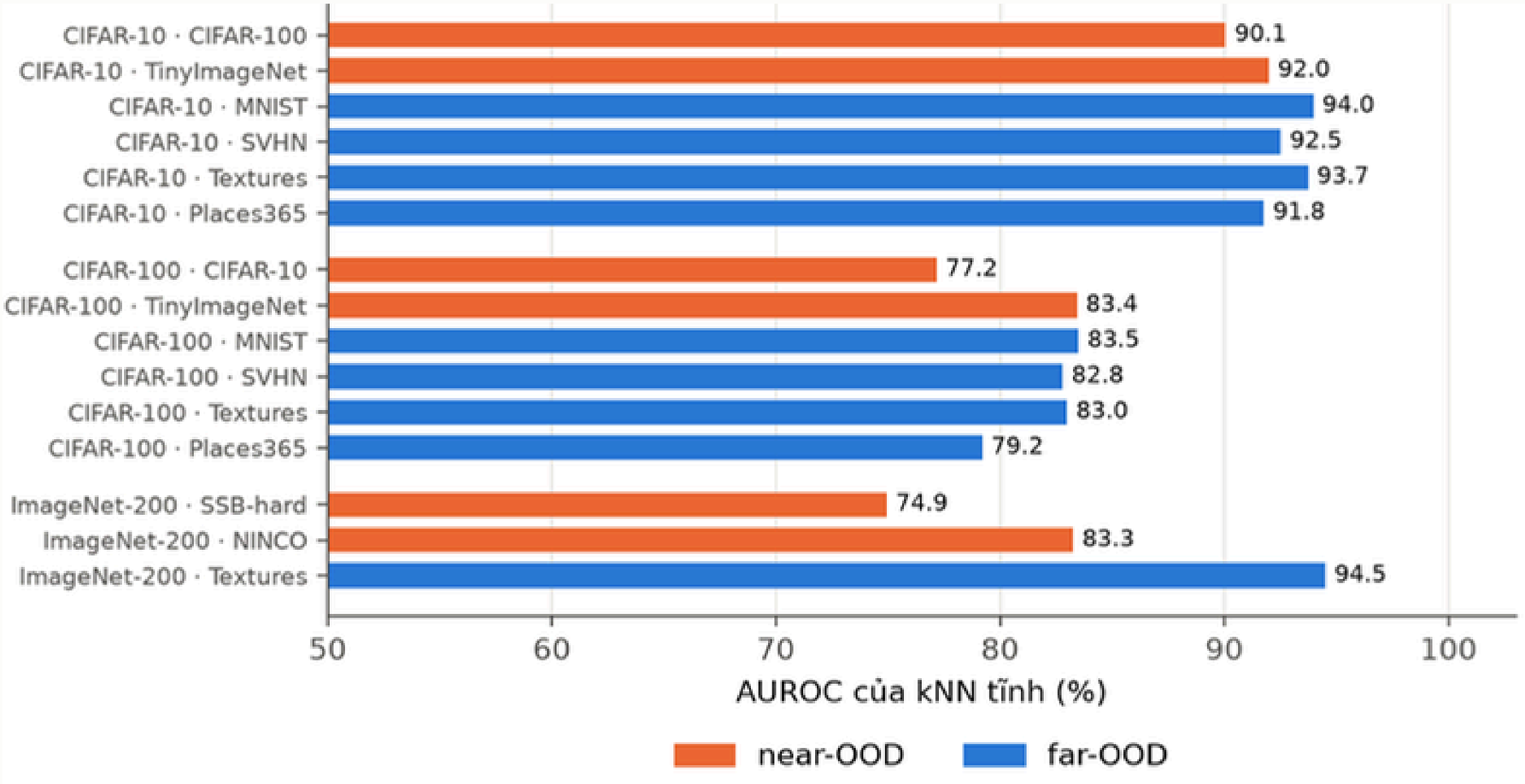
6.3 EDA: Sample Images (ImageNet-200)

SSB-hard and NINCO are other ImageNet classes in the same photographic style; far-OOD is Textures. SSB-hard is the hardest set in the benchmark (kNN AUROC 74.9).



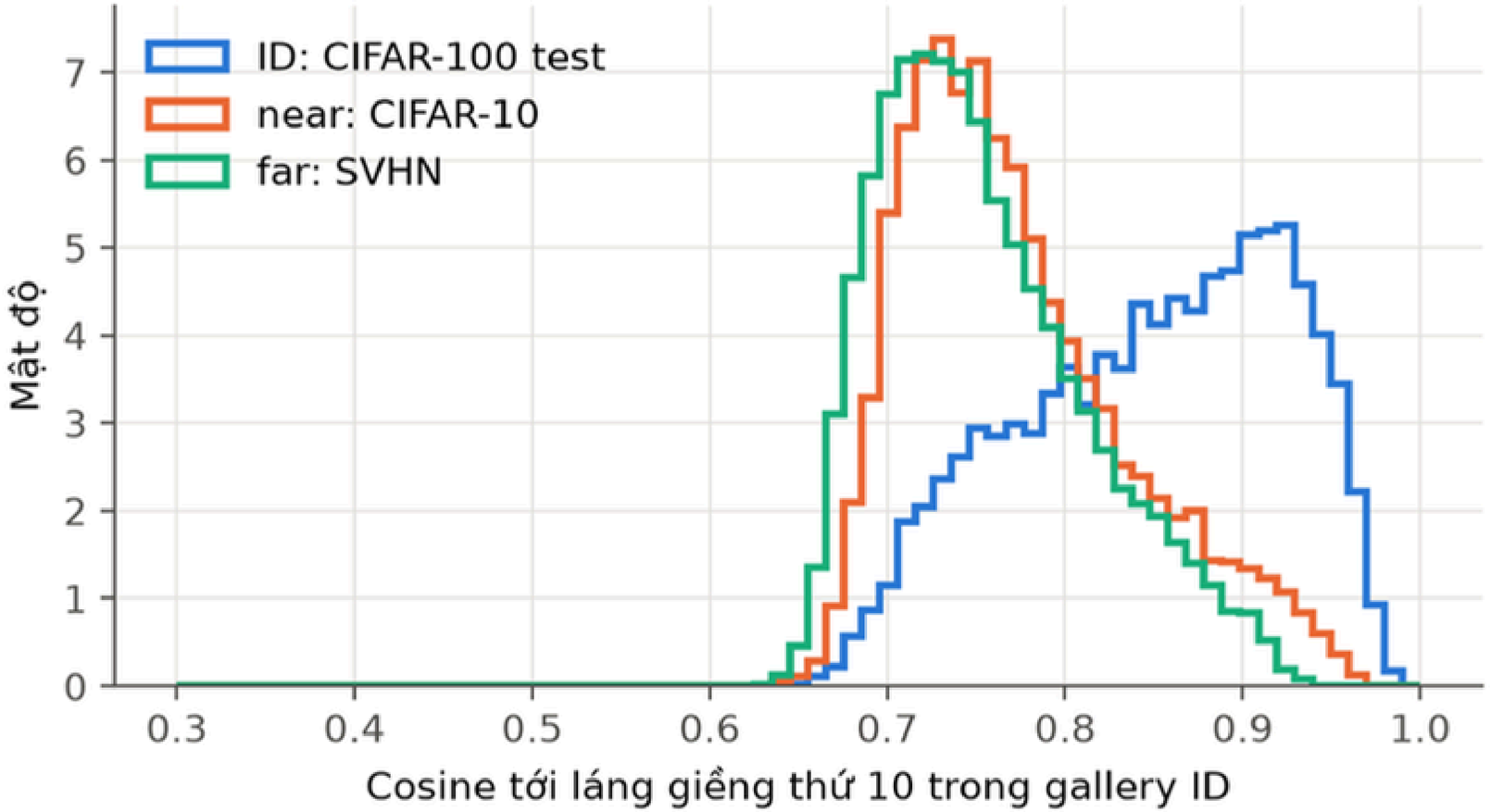
6.3 EDA: Difficulty

Static kNN AUROC: 90–94 on CIFAR-10, only 77–84 on CIFAR-100 and 74.9 on SSB-hard. This is where memory has room to help.



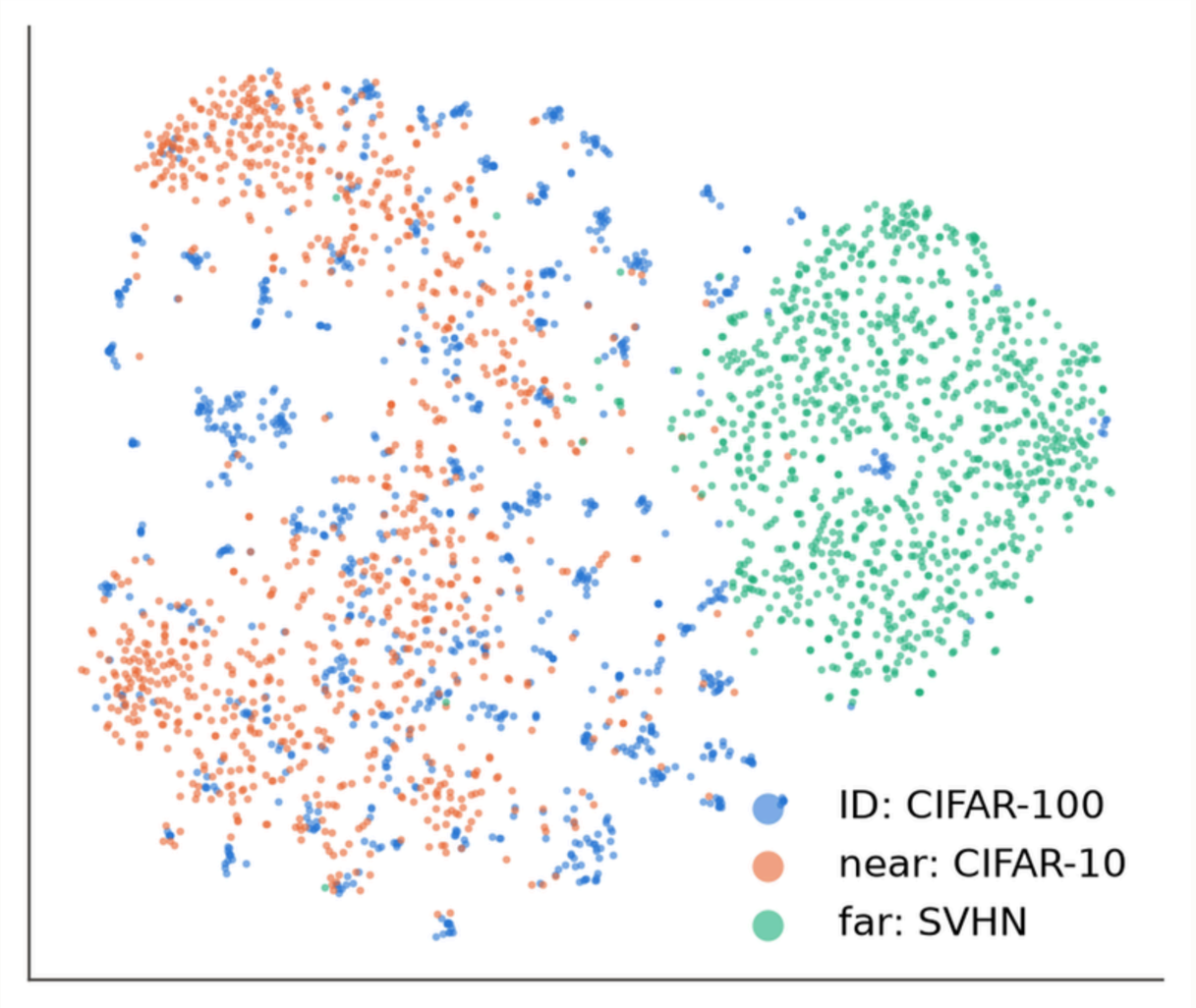
6.3 EDA: Limits of the Static Score

On CIFAR-100, kNN scores of CIFAR-10 (near) and SVHN (far) almost coincide and overlap the ID tail. Distance to ID cannot see that OOD samples resemble each other.



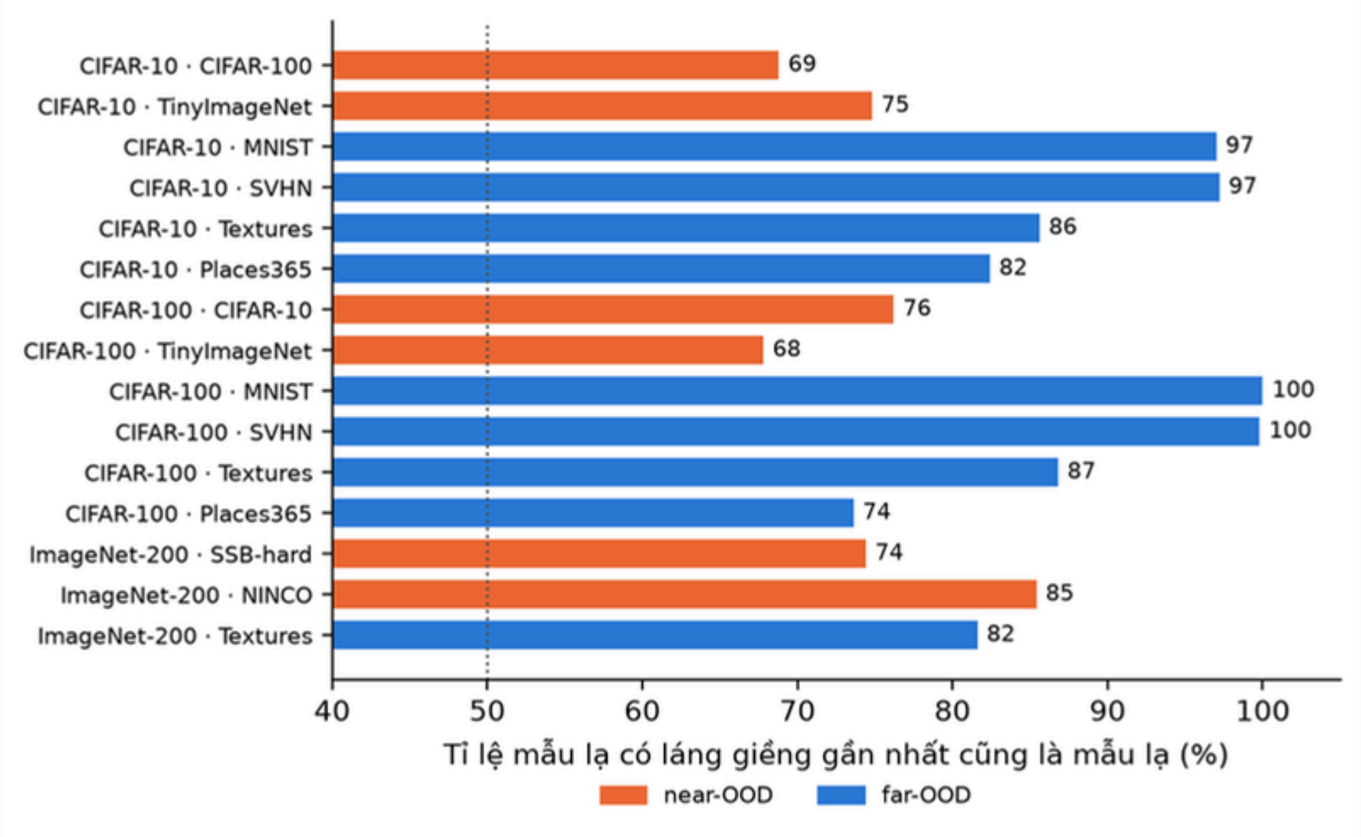
6.3 EDA: Feature-Space Structure

t-SNE of ResNet18 features: SVHN forms its own cluster, far from ID; CIFAR-10 mixes with CIFAR-100 but keeps its own dense regions.



6.3 EDA: Do OOD Samples Cluster?

Share of OOD samples whose nearest neighbor is also OOD (500 OOD + 500 ID; chance = 50%): far-OOD 97–100%, near-OOD 68–85%. Memory should help most on far-OOD.



6.3 EDA: ID Drift (CIFAR-10-C)

Gaussian noise (level 5) looks almost as OOD as real near-OOD to static kNN (88.5 vs 91.0); brightness (level 3) barely drifts (52.2). Without labels, drifted ID and true OOD cannot be told apart.

